Tracking the Green Transition

A Data Science Methodology to Identify the UK Green Workforce

Department for Energy Security and Net Zero (DESNZ)

1. The Raw Material: Millions of Job Adverts

The Dataset Scale

To accurately measure green skills, we cannot rely on broad industry codes. We need to read the actual tasks employers are requesting.

We sourced **62.3 million UK job advertisements** spanning a 6-year period. This represents the vast majority of online labor demand in the UK economy.

The Cleaning Process

Standardization: Text converted to lowercase; grammar, special characters, and "stop words" removed.

Noun Chunk Extraction: Rather than isolating single words, the NLP extracts full technical concepts (e.g., "offshore wind turbine" instead of just "wind").

Output: Billions of raw, unstructured phrases ready for mathematical filtration.

2. Phase 1: Semantic Filtering

How does an AI know a phrase is "Green"? We convert every extracted phrase into a 300-dimensional mathematical vector and calculate its distance to the core concept of **"environment"**.

$$ \text{Cosine Similarity} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} $$

The 0.35 Threshold

We deliberately set a precisely calibrated **0.35 similarity threshold**. Any phrase scoring below this is permanently deleted from the pipeline.

Why it matters

This mathematically captures functional green technology (e.g., "photovoltaic cells") while instantly discarding unrelated office jargon (e.g., "team catering").

3. The "Generic Buzzword" Problem

Semantics are not enough.

The phrase *"paper recycling"* is highly semantically related to the environment (it passes Phase 1). However, almost every standard office worker uses a recycling bin.

If we use semantics alone, we will capture standard administrative workers and artificially inflate the size of the Green Economy to include the entire country.

- ✗ **Generic:** "Recycling Bin" (Used by everyone)
- ✗ **Generic:** "Friendly Environment" (Context error)
- ✓ **Specialized:** "Decarbonisation Strategy"
- ✓ **Specialized:** "Carbon Auditing"

4. Phase 2: Relative Share Mathematics

To solve the buzzword problem, we introduced statistical rigor. We established "Anchor SOCs" (e.g., Wind Energy Engineers, Environmental Scientists) representing undeniable green workers.

$$ \text{Relative Share} = \frac{\% \text{ of Anchor Adverts Using the Term}}{\% \text{ of Total Economy Adverts Using the Term}} \geq \mathbf{10.0} $$

> **The 10x Rule:** A phrase is only added to our Green Dictionary if it is used **at least 10 times more frequently** by green professionals than by the average UK worker.

> **The Result:** "Paper recycling" fails the test. "Carbon sequestration" passes. Greenwashing is statistically eliminated.

5. Phase 2.5: The Gold Standard Polish

The Final Sanity Check

Even with the 10x multiplier, anomalies can survive. To ensure absolute precision, we apply a final, strict semantic polish.

The 0.45 Pillar Test

Every surviving phrase is checked against a strict list of undeniable pillars, such as **"Net Zero"** and **"Renewable Energy"**.

$$ \text{Similarity to Pillars} \geq 0.45 $$

If a phrase fails to meet this 0.45 threshold, it is permanently discarded. This leaves us with a mathematically verified, pristine Green Dictionary.

6. Phase 3: Classification via The Rule of 3

Scanning the Economy

With our verified dictionary built, we run all 62.3 million original job advertisements back through the keyword scanner.

The Classification Gate:

$$ \geq \text{3 Distinct Dictionary Terms} $$

Why 3 distinct terms?

Many companies place standard boilerplate text at the bottom of adverts ("We are committed to Net Zero and Sustainability").

Requiring 3 *different* technical terms proves the environmental tasks are a core, operational requirement of the role, not just HR marketing.

Results: Green Jobs Over Time

Tracking the absolute volume of validated Green Jobs identified by the NLP pipeline across the 6-year period.

[Insert Line Chart: Count of Green Jobs Over Time Here]

Results: Green Jobs by Occupation

The distribution of identified Green Jobs across standard Occupational (SOC) classifications, highlighting the spread of green tasks beyond traditional sectors.

[Insert Bar Chart / Table: Count of Green Jobs by SOC Code Here]